IC103 Lecture 2: Key Takeaways

Elasticity and Consumer Choice

1. Price Elasticity of Demand

- **Elasticity** measures how responsive quantity demanded is to a change in price.
- **PED =** % change in quantity demanded / % change in price.
- **PED > 1:** elastic.
- **PED < 1:** inelastic.
- **PED = 1:** unit elastic.



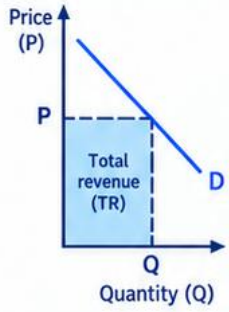
The larger the absolute value of PED, the more responsive quantity demanded is to price changes.

2. What Affects Elasticity?

1. Demand is more elastic when there are **close substitutes**.
2. **Necessities** tend to be more **inelastic**.
3. **Luxuries** tend to be more **elastic**.
4. The easier it is to **switch away** from a good, the more **elastic** demand tends to be.

3. Elasticity and Revenue

- **Total expenditure (TE)** = price × quantity.
- In this simple setting, **TE = TR = P × Q.**
- If demand is **elastic,** price and total revenue move in opposite directions.
- If demand is **inelastic,** price and total revenue move in the **same** direction.
- If demand is **unit elastic,** total revenue stays unchanged.



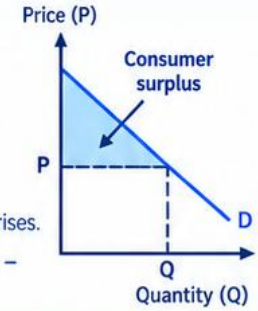
Total revenue is the area of the rectangle under the demand curve at the given price and quantity (TR = P × Q).

4. Other Elasticities

1. **Income elasticity of demand:** response of demand to a change in income.
2. **Cross-price elasticity of demand:** response of demand for one good to a change in another good's price.
3. **Price elasticity of supply:** response of quantity supplied to a change in price.
4. **Normal goods** usually have positive income elasticity; **inferior goods** have negative income elasticity.

5. Utility and Consumer Surplus

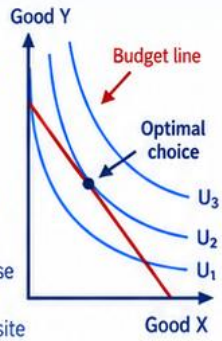
- **Utility =** satisfaction from consumption.
- **Total utility (TU):** satisfaction from all units consumed.
- **Marginal utility (MU):** extra satisfaction from one more unit.
- **Diminishing marginal utility:** extra satisfaction usually falls as consumption rises.
- **Consumer surplus =** willingness to pay – price actually paid.



Consumer surplus is the area between the demand curve and the market price, up to the quantity consumed.

6. Consumer Choice

- **Indifference curves** show combinations of two goods that give the same satisfaction.
- The **budget line** shows what the consumer can afford.
- The consumer chooses the highest attainable indifference curve.
- A price change affects choice through the substitution effect and the income effect.
- For a normal good, both effects of a price rise reduce quantity demanded; for an inferior good, the income effect works in the opposite direction.



Quick Revision Checklist

- ☐ Define PED.
- ☐ Explain what makes demand elastic or inelastic.
- ☐ Explain the link between elasticity and revenue.
- ☐ Define utility, TU and MU.
- ☐ Explain the substitution and income effects.